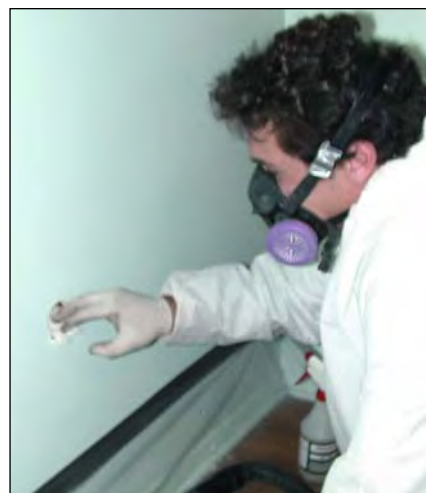




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Lead-Safe Weatherization

With a little inventiveness and a little help from their friends, a weatherization agency in Denver has learned to provide energy efficiency measures—sawing a lot of holes in walls painted with lead paint—without endangering the lives of its workers.

by Charles Wright



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Housing in Denver, Colorado, like housing in most large cities, is a mix of leaded and lead-free homes. The good news is that there was less lead used here in the West than in the East—15% of homes contaminated in the West, compared to 40% in the Northeast, according to a 1999 HUD housing survey. The bad news is that we're not lead-free. The low-income homes that provide great opportunities for energy conservation also have lead paint contamination blocking the way for window

wear and tear, and to temperature and weather changes, and they resisted mold and mildew in moist areas. From about 1950 to 1978, the concentration of lead in paint was reduced, as other pigments became more popular. Lead-based paint was banned from residential use in 1978, since by that time the health risks from lead exposure were well-known. Today, the legacy of lead-based paint is the hazard of lead-contaminated dust and debris. Pregnant women and young children are especially at risk. Children

and develop improved treatments for lead exposure.

Our Rationale

To fully comply with both the required assessment and testing, and—where tests for lead are positive—the notification, remediation, and abatement that OSHA, EPA, and/or HUD require, we would have to devote a disproportionate amount of our resources to 15% of our homes, at the



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(left) Benito peels tape and plastic from a wall as clean-up proceeds. Notice on the plastic how little debris there is to be handled. (right) A shrouded drill, HEPA vacuum, and simple cleaning supplies are quickly assembled and ready for use.

replacement and reglazing; various ducting, venting, and other repairs; and—most critical and cost-effective of all—insulation. Sun Power, the company I work for, does few window replacements and usually subcontracts that work and other repairs to qualified contractors. But lead-safe insulation work is our challenge. And so is answering this question:

With nearly 1,000 homes to weatherize in the year ahead, how can we deliver a quality energy conservation service, protect staff and customers, and meet all of our production commitments at the same time?

A Little History

As far back as 1884, lead was added to paint to improve durability and for pigment, especially in light-colored paints. Lead-based paints stood up to

under the age of six are at risk of poisoning from very small amounts of lead, and because their brains and nervous systems are still developing, lead poisoning can cause irreversible damage.

The Occupational Safety & Health Administration (OSHA) has established ample regulations to protect workers from the hazards of lead. The EPA has developed extensive procedures for the construction trades, covering a wide range of remodeling and renovation work. These procedures are designed largely to protect the residents from harm, while minimizing risk to the workers. HUD has established standards to address the problem of lead in the home; these are generally consistent with the EPA standards as applied to residences. The Centers for Disease Control and Prevention (CDC) promote state and local screening efforts

cost of reducing the effectiveness of our mission—energy conservation in low-income housing.

Sun Power is the largest nonprofit low-income weatherization provider in our region, with nearly 1,000 homes to weatherize on schedule this year. One out of two of these homes need wall or other closed-cavity insulation. We install insulation almost exclusively by drilling through the interior surfaces of the exterior ceilings and walls and then blowing in cellulose insulation and dense-packing it into the wall cavities. We do this now with a new understanding of lead hazard health and safety issues. We have a certified lead risk assessor on staff and a working partnership with our local Healthy Homes program for lead abatement. (HUD's Healthy Homes Initiative is a nationwide effort to research and demonstrate low-cost home hazard

assessment and intervention methods. Along with lead safety, the initiative promotes structural, electrical, and fire safety in the home. See the end of this article for more on the Healthy Homes Initiative.)

But after swimming through the sea of regulations and conferring with state and federal officials, we determined that we do not do lead abatement, remodeling, construction or repair, and we do not in our weatherization work disturb more than 2 ft² per room or 20 ft² of the total exterior surface of a house. We aspire to DOE's mandate to train, equip, and supervise our staff to accomplish lead-safe weatherization. We recognize that a significant fraction of the homes we work in may pose a potential lead paint hazard. Additionally, from time to time we weatherize HUD-funded properties, and when we do so we must meet HUD's mandated guidelines for lead-safe work practices.

We did not want lead issues to distract us from our mission of providing services to people who would otherwise do without. In order to reconcile our energy conservation mission with the caveat to "do no harm," we needed a solution that would facilitate the former and include the latter, without compromise.

We wanted to know how big a lead problem we really had. The best way to determine if lead poisoning is present is to take a blood lead level test. So we asked our contract occupational medicine provider to test the blood lead levels of our full field staff in order to establish a baseline and to assess the degree of jeopardy to which our people had already been exposed. Like most weatherization providers, we have experienced people with nearly two decades in the field, and new hires still in training. Had we unknowingly exposed

staff to lead hazards resulting in ever-higher lead concentrations in our tried-and-true personnel?

When the test results were delivered to us, the data held two surprises. All but two of our people tested normal for the presence of lead, regardless of their years of

able gloves, and HEPA-rated respirators, along with the usual kneepads, hearing protection, hard hats, safety glasses, first aid, and eyewash stations. So by adding HEPA-rated vacuums and wash-up stations to our field equipment, and by training our people to use them



A finished drill shroud lies on Joe Hall's bench with some of the component parts and basic tools needed to make this effective device.

service. The two who tested high had not been in the field more than a year or two, and each had a clear explanation for his high blood lead level that had nothing to do with weatherization. One had worked for years in a lead/acid battery factory and the other carried a bullet in his arm (not an on-the-job injury)! While we were reassured by the blood tests, we decided to step up our worker safety program to ensure that staff would remain safe.

For the last decade and a half, Sun Power has provided its field staff with disposable coveralls, two types of dispos-

properly, we can continue to do an excellent job of protecting our staff.

Protecting our customers was a different challenge. We have routinely used 2-mil plastic to protect our clients' possessions and facilitate cleanup. Also, we have been cautioning residents about dust and providing them with the EPA booklet *Protect Your Family from Lead in Your Home*, largely to minimize any potential respiratory health or hazardous materials problems. Our auditors and crew leaders routinely discuss with the residents any actual or potential hazards observed dur-

A Further Innovation

I recently received an e-mail about another variation of the drill shroud by Terry Lauer, an inspector/auditor of the Weatherization Assistance program with the Northeastern Colorado Association of Local Governments, that is

both elegant and practical (see photo on page 27). Rather than attach the shroud to the drill, Terry used a bearing to mount it directly on the shaft of a short drill extension, thereby enabling it to fit virtually any drill. This design also makes the tool simpler to use in the field, lighter, and more durable.

For more information on this approach to the drill shroud, contact H.J. Greenwood at the Northeastern Colorado Association of Local Governments, 231 Main St, Ste. 211, Fort Morgan, CO 80701, or by e-mail at gerwoodhj@twol.com.

ing the initial home assessment, and they suggest basic precautions that residents can take to reduce those hazards. We checked our insurance to be sure that we were covered for lead hazards, without exclusions. But to reduce or eliminate all lead paint dust risk to our customers would require us to rethink our process and field-test new procedures.

The Solution

Early in our field testing, Pete Will, our quality assurance and training supervisor and our safety officer, tried a paint stripper designed to encapsulate layers of paint and bind them in a rubbery compound. The paint stripper worked only moderately well, and it had to set overnight to cure, making it necessary to disturb our clients by coming back the next day. In the end, we solved the problem thanks to Joe Hall at Longs Peak Energy Conservation in Boulder, Colorado. Like all of the eight weatherization providers in the state, Longs Peak was looking for answers, and Joe provided us all with the cornerstone of cost-effective lead safety compliance.

As all those who have wrestled with these thorny issues learn, the key is containment. Containment may mean many things, from the gooseneck seal on a garbage bag to wrapping a house in enough plastic to make the conceptual artist Christo proud. To Joe it meant thinking small and addressing the potentially hazardous paint dust generated from drilling through old plaster and drywall at the source. Joe searched the Internet for tools and found a commercial shroud designed to fit over a drill and catch this dust. The dust would then be vacuumed out through an exhaust fitting on the side. Good idea for abatement, but too expensive for a weatherization agency living on a budget.

Joe's solution was to create an affordable drill shroud that could be built so quickly and cheaply in the back room that we could give one to each of the auditors and field crews. Joe taught us to assemble half of a plastic accordion toilet

plunger, a short length (3 or 4 inch diameter, depending on the size of the drill) of white PVC pipe, a 1 1/2-inch PVC elbow, and some miscellaneous glue and clamps. The end product was simple, but it was a thing of practical beauty to us. The shroud was clamped to the business end of a drill and hole saw and attached to a HEPA vacuum that

his work. We also found that not all of our various brands and configurations of drills and saws fit the shroud. It would be an additional expense, but we replace old tools on a rotating basis anyway and we had a workable budget for power tools, so the new criterion was that all our new drills and saws must fit the shroud.

A Day in the Life

The whole process is straightforward once the customer makes the commitment to insulate. We explain the hazards of working with lead and describe the lead-safe weatherization process. When the customer is comfortable with the procedure, the area including the walls to be treated is cleared of furniture, hangings, and extraneous people. Then it is draped with 5-ft-wide drop cloth of 2-mil plastic, secured about 1 ft up the wall by a continuous run of blue tape. Our technicians suit up in their usual disposable coverall, rubber gloves, and HEPA-rated respirator along with kneepads, hearing protection, and safety glasses, as needed.

When the room and its contents are prepared, the HEPA vacuum, drill, and shroud are assembled and checked. The holes are sawed, the dust is sucked into the vacuum, and the divot of each holesaw cutout is dropped directly into the wall cavity. When all of the holes in one room have been sawed, the assembled wall tool is moved to the next room or disassembled and cleaned over the plastic drop cloth. Each hole is wiped down with a moist paper towel, which is also dropped into the cavity. The paper towel, like the divot, could be removed with the other dust and debris caught by the plastic, but because the shroud makes the process so clean, our crews often prefer simply to encapsulate them in the wall cavity, rather than having them underfoot to add to the cleanup problem.

The plastic may be removed and replaced if it gets dirty, but it usually stays in place until the insulation blow for that room is complete. Sometimes a



A technician demonstrates Terry Lauer's evolved version of the drill shroud on a lead-free surface.

could be worn, carried, or dragged behind the operator. Because the hole saw slightly recessed into the accordion plunger, you cannot saw a hole without the plunger coming into firm contact with the wall or ceiling, so that all of the dust is contained within a 4-inch diameter of the hole. The tool worked so well that it became integral to the success of our lead paint health and safety program. (See "A Further Innovation" for the next-generation drill shroud!)

But we need to do more than just drill holes. From time to time we have to cut in an attic or kneewall access with a reciprocating saw. Benito Velasquez of Sun Power found that revising the drill shroud to serve on a saw meant replacing the large-diameter PVC pipe with a clear Plexiglas tube. Benito found that the clear tube quickly became coated with dust, obscuring the blade. But coating the interior of the tube with a spray intended for saw tables prevents dust from sticking to the surface, allowing the installer to see

Weatherization

crew will install two layers of plastic while prepping the room. After the holesaw work is done, the top layer is removed, and the second layer is left for the insulation cleanup. When the room is insulated, plugged, and patched, cleanup proceeds quickly. The crew mist down any dusty areas; then they roll up the plastic and store it in heavy 3-mil trash bags for disposal. Potentially hazardous dust is kept well below the EPA action levels. The whole process adds perhaps one to two hours to a full insulation package for a crew of two or three, and it's a practical and effective means of ensuring a lead-safe work environment and a healthy and efficient dwelling place.



Charles Wright is a 16-year veteran of residential energy conservation. He is currently consulting under contract to Sun Power, Incorporated, in Denver, Colorado.

For more information:

HUD's office of Healthy Homes and Lead Hazard Control offers resources and information for homeowners and home performance professionals on lead safety. Go to their Web site, www.hud.gov/offices/lead.

For more information on HUD's Healthy Homes Initiative, go to www.hud.gov/offices/lead/hhi/index.cfm.

For a complete parts list and instructions on how to build your own shrouds, or for Sun Power's specific Lead Safe Practice working protocols incorporating the shrouds, contact:

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Please let us know how the shrouds work for you and how the tool evolves in your operations. Contact us at Sun Power, as above, or e-mail [Charles Wright at csquarew@concentric.net](mailto:csquarew@concentric.net) or call him at (303)860-7786.